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Effect of copper on highly effective Fe-Mn based catalysts during production of light olefins *via* Fischer-Tropsch process with low CO_2 emission

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ABSTRACT

The effect of Cu on Fe-Mn based catalysts system in production of light olefins *via* Fischer-Tropsch process was investigated. Catalysts with different Cu loadings were synthesized and evaluated. The 3.0 wt% Cu promoted Fe-Mn catalyst demonstrated an activity as high as 3.58×10^{-6} mol_{CO} g_{cat}^{-1} s^{-1}. The corresponding conversion of CO and selectivity to light olefins are 96.9 % and 40.1 %, respectively. Characterization results indicated that the Cu promoter facilitated the reduction process and enhanced the H-assisted CO dissociative adsorption, by changing the interactions between metals and the SiO_2, which lead to an improved activity. The overall weakened surface basicity caused by Cu decreased the chain growth probability and lead to a higher selectivity to light olefins by shifting the product distribution toward short-chain hydrocarbons. Furthermore, all prepared catalysts showed superior efficiency of carbon utilization with CO_2 selectivity as low as 23.0 % due to the hindered water-gas shift (WGS) reaction.

1. Introduction

Light olefins (*e.g.*, ethylene, propylene, and butylene) are extensively used chemical building blocks in chemical industries for producing a wide range of products (*e.g.*, polymers, plastics, synthetic textiles, solvents, and coatings). Traditionally, light olefins are manufactured through petroleum-based processes including catalytic cracking of naphtha, dehydrogenation of saturated hydrocarbons or as byproducts of crude oil refining. However, the contradiction between the growing demand of light olefins worldwide and the rapid depleting of limited oil sources is becoming acute. In light of environmental concerns, alternative to petroleum-based feedstocks for light olefins has attracted greater attention. Syngas ($H_2 + CO$) conversion is of interest because it can be directly derived from steam reforming of natural gas or gasification of renewable biomass. In general, there are two different approaches to convert syngas into light olefins: indirectly *via* methanol or dimethyl ether (DME) synthesis followed by methanol/DME to olefin (MTO/DMTO, respectively), or directly either *via* Fischer-Tropsch to olefins (FTO) or *via* combined catalytic process. The direct process is considered more attractive from an environmental and economic point of view [1–7].

The direct process can be further categorized according to the reaction mechanisms. One category is defined by the use of bi-functional catalysts to combine multiple catalytic steps (*e.g.*, DME synthesis and DMTO, or methanol synthesis and MTO). High selectivities to light olefins have been achieved on these catalysts. Bi-functional ZnCrOx/SAPO and Zr-Zn/SAPO-34 catalysts achieved light olefin selectivity up to 80 % and 63 % at CO conversion of 17 % and 9.5 %, respectively [2,8]. These catalysts generally have comparatively low CO conversion due to the competition between the optimization of reaction conditions for the different active phases [9]. The second category is based on the modification of conventional FT catalysts to maximize the selectivity to

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light olefins by appropriately restricting the chain growth reaction. A 52 % selectivity to light olefins at the CO conversion of 88 % was achieved by Torres Galvis et.al when using Fe/CNF nanoparticle catalyst [1]. Zhang et.al reported that the Ag promoted Fe_3O_4 catalysts with microporous structure achieved a light olefins selectivity of 43 % at CO conversion of 71.4 % by carefully tuning the pore size and Fe dispersion [10]. However, both strategies tend to generate high percentage of CO_2. For example, the CO_2 selectivity was higher than 40 % on the MnO_x/MSAPO catalysts at 400 °C and amounted to 47 % on a CoMn catalyst with nanoprisms structure [3,11]. Wang et al. [12] estimated the annual energy consumption as a function of CO_2 selectivity by modeling a typical FT plant through Aspen Plus 7.2 and showed there would be approximately 20 % more energy consumed annually on compressing and heating if the CO_2 selectivity increased from 20 % to 40 %. Therefore, increasing carbon utilization efficiency in those processes by tuning the selectivities towards light olefins and away from CO_2 is highly necessary [13,14].

Iron-based catalysts occupy an indispensable position due to their low cost, flexibility to the H_2/CO ratio, as well as high light olefins selectivity at relatively high reaction temperature required by FTO process [7,15]. The product distribution highly depends on the promoter and catalyst structure, which shows the possibility to maximize the carbon utilization efficiency by investigating different types of Fe-based catalysts. Zirconium and Zinc were reported as very good structural promoters for Fe-based catalyst that can reduce the CO_2 generation and improve the light olefins selectivity at high CO conversion [16,17]. Mg and K dual-decorated Fe/rGo catalysts were also reported for FTO process with mitigated CO_2 emission and enhanced catalytic activity by inhibiting the WGS reaction [13]. As low as 15 % of CO_2 selectivity was achieved at CO conversion of 95.1 % and light olefins selectivity of 53.3 % over a FeZn catalyst when co-feeding 8% of CO_2 in the syngas [4]. Copper has been proposed and widely accepted as a reduction promoter, which can decrease the required reduction temperature and ease the reduction process. However, the effect of Cu on the product distribution during an FTO process has not been clearly addressed. Zhang et al. [18] reported that copper has strong influence on the product distribution by shifting the product towards heavy hydrocarbons with enhanced olefin/paraffin ratio, while Li et al. [19] and Wan et al. [20] observed that copper promoted iron-based catalysts have higher selectivity to light hydrocarbons including CH_4 and higher paraffin content. The controversy exists because the FT process is complicated and usually conducted under different reaction conditions on different catalyst systems.

In this study, Fe-Mn based catalysts with different Cu loadings were prepared and the enhanced light olefins selectivity is demonstrated at CO conversions above 90 % with CO_2 selectivity below 25 %. The effects of Cu on the product distribution are discussed. Various characterization techniques including Brunauer-Emmett-Teller (BET) surface area, inductively coupled plasma mass spectrometry (ICP-MS), X-ray diffraction (XRD), scanning electron microscope (SEM), transmission electron microscope (TEM) with energy dispersive X-ray spectroscopy (EDX) mapping, H_2-temperature programmed reduction (TPR), CO_2/H_2/CO temperature programmed desorption (TPD), H_2-derivative thermogravimetry (DTG) as well as *in-situ* diffuse reflectance infrared Fourier transform spectroscopy (DRITFS) were used to investigate the structure-performance relationships of the catalysts.

2. Experimental

2.1. Catalyst preparation

Catalysts with different loading amounts of Cu were prepared by combining co-precipitation and impregnation method. Aqueous HNO_3/NH_4OH, pH = 9.2 (± 0.2), was introduced into a precipitation vessel with silica gel and pre-heated to 70 °C. Aqueous $Fe(NO_3)_3$ and $Mn(NO_3)_2$ (weight ratio Fe/Mn/SiO_2 = 100/10/12) was added into the precipitation vessel. An additional amount of NH_4OH solution was added simultaneously to maintain the pH at a constant value of 9.0 (± 0.2). The temperature of the precipitation vessel was kept at 70 °C during the whole precipitation process. After precipitation, the obtained slurry was aged under vigorous stirring for 6 h. The precipitate was filtered, dried at 90 °C, and ground to 120 mesh. The appropriate amount of copper promoter was introduced by impregnation of aqueous $Cu(NO_3)_2$. The obtained catalyst precursors were further dried at 90 °C for 4 h and calcined under static air at 550 °C for 6 h. All samples were crushed and sieved to 120 mesh before further characterization and evaluation. The designed weight ratios of Fe/Mn/Cu/SiO_2 are 100/10/x/12 (x = 0, 0.5, 1.5, 3), and the samples are labeled as Cu-0, Cu-0.5, Cu-1.5 and Cu-3.0, accordingly.

2.2. Catalyst characterization

2.2.1. Textural properties and composition

The Brunauer-Emmet-Teller (BET) surface area, average pore size of the catalyst, total pore volume as well as the Barrett-Joyner-Halenda (BJH) pore size distribution were measured *via* N_2 adsorption/desorption at −196 °C using a gas sorption analyzer (Quantachrome, Quadrasorb EVO). Each sample was degassed at 250 °C under vacuum for 10 h prior to analysis. The ICP-MS (PerkinElmer NexIon 350b) was used to verify the content of Fe, Mn, K, Cu, and SiO_2 in the catalyst samples. Each sample was mixed with Lithium metaborate ($LiBO_2$) and fused at 1100 °C for 5 min under static air atmosphere. The collected glass-like mixture was then dissolved into 5 wt% nitric acid and washed 5 times (stirred for 20 min for each iteration). The solution was diluted further to get the testable sample for ICP-MS.

2.2.2. Structure and morphology

XRD patterns were measured by a Rigaku Smart-lab X-ray diffraction system using Cu-Kα1 (λ = 0.15406 Å) radiation (40 kV, 40 mA) to verify the crystal structure on the calcinated and spent catalyst samples. The scanning range (2θ) was from 10° to 90° at a scanning rate of 4°/min and chopper increment of 0.02°. The morphology and elements dispersion of the catalysts were studied by SEM (FEI, Quanta FEG MK2) and TEM (FEI, 80-300 Titan-S) with energy dispersive X-ray spectroscopy (EDX) mapping.

2.2.3. Fourier-transform infrared spectroscopy (FTIR)

In situ FTIR spectra were collected using a Nicolet iS50 FT-IR spectrometer equipped with the Harrick Praying Mantis diffuse reflection environmental chamber (HVC-DRM-5). The dome shape sample cell with 4 mm ZnSe window was connected with a gas feeding system to allow *in-situ* measurements during the FTO process. Spectra were collected at 64 scans with 4 cm^{-1} resolution.

Prior to the *in-situ* FTIR tests, catalyst samples were mixed with KBr (KBr/Sample = 10:1, weight ratio) and ground for better signal. Samples were reduced at 20 bar while heating to 500 °C at 10 °C/min under flowing H_2 and maintained for 2 h. The pressure was then released, while samples were treated at 500 °C under atmospheric pressure and He flow for 30 min before cooling down to room temperature. Background signal was collected at 20 bar under flowing H_2 and CO (H_2/CO = 2.0). IR spectra were collected continuously while the system was heated to 300 °C (10 °C/min) and maintained for 2 h. The total flow rate of feed gases was maintained at 10 ml/min by mass flow controllers (MFCs).

2.2.4. Temperature programmed analysis

In the temperature programmed process, all gases were introduced into a purification system prior to sample cells. The gas flows were maintained at 50 ml/min for TPR and TPD experiments, and 100 ml/min for the DTG analysis. The temperature ramp was set at 10 °C/min in all cases.

The H_2-TPR experiments were performed using a chemisorption analyzer (Quantachrome, Autosorb IQ). Typically, 100 mg of catalyst was loaded into a quartz sample tube (10 mm i.d.). A flow of 5 % H_2 in He was used as reduction gas. The effluent gas was analyzed by a thermal conductivity detector (TCD) to monitor the concentration variation of H_2. The catalyst was then reduced under H_2 with the temperature increasing from 100 to 900 °C.

The H_2/CO/CO_2-TPD experiments were performed using the same system as for TPR tests, and all samples were reduced *in-situ* under H_2 prior to the adsorption and desorption process. 100 mg of catalyst was loaded into the sample cell and the temperature was kept at 150 °C with He flow for 60 min to remove the adsorbed species on the catalyst. Then the gas was switched to H_2 and the temperature was increased to 500 °C for the 2 -h reduction. After the *in-situ* reduction, the system temperature was cooled down to 50 °C. The H_2/CO/CO_2 adsorption process was then performed at 50 °C for 90 min, after which, the catalyst was purged with He for another 60 min to remove the weakly adsorbed H_2/CO/CO_2. Next, the H_2/CO/CO_2 desorption behaviors were recorded with the temperature increasing from 50 to 900 °C.

The DTG experiments were carried out using a thermal analyzer (TA instrument, Q600). Typically, a 50 mg catalyst sample was loaded into a sample cell per test. A flow of 10 % H_2 in He and 10 % O_2 in He were used as reduction and oxidation gases, respectively. The weight variation of the catalyst sample was recorded during the following program. Firstly, the catalyst was reduced by H_2 flow with the temperature raising from room temperature to 1000 °C at 5 °C/min and hold for 30 min. After the reduction process, He was purged into the sample cell until the sample was cooled down to room temperature. Then, the O_2 flow was introduced and the catalyst was oxidized with the same temperature program. Finally, the oxidized sample was reduced again using the same procedure as in the first step.

2.3. Catalytic tests

The FTO performance of the catalysts was examined in a down-flow stainless fixed-bed reactor with an inner diameter of 9 mm (Fig. S1). Typically, the catalyst was loaded into the reactor with a K-type thermocouple inserted in the middle of the catalyst bed (Fig. S1, label 10–13). The reactor was put into a furnace equipped with a temperature controller. The flow rate of H_2 and CO were controlled by two MFCs (Fig. S1, label 5–6, Parker Porter Mass Flow, Series 201). The outlet of the reactor was connected with a hot trap (Fig. S1, label 16, 200 °C) and a cold trap (Fig. S1, label 17, 2 °C) to collect the wax and liquid products and then a back-pressure regulator (Fig. S1, label 18) to maintain pressure. The effluent gas flowed into a wet type gas flowmeter (Fig. S1, label 19, Ritter TG 05/3) to record the real-time flow rate and accumulated volume before gas chromatography (GC). The effluent gas was analyzed by an online GC (Fig. S1, label 20, model 8610C-MG#5, SRI) equipped with a flame ionization detector (FID) and a thermal conductivity detector (TCD). Two sample gas routes controlled by a 10-port sampling valve inside the GC were employed to analyze the hydrocarbons, H_2, CH_4, CO and CO_2 with a single injection. The first gas route was for the FID channel, where the hydrocarbons were separated in a capillary column (HP-AL/S, 60 m × 0.53 mm). The other route was for the TCD channel, a HeySep D column (3 ft × 1/8 in) was used as a guard column and H_2, CH_4, CO and CO_2 were separated in a ShinCarbon column (ShinCarbon, 2 m × 1/8 in) before analyzed by the TCD. The results from the two channels were linked and normalized using CH_4 as the bridge since CH_4 can be well quantified in both TCD and FID channels. The liquid products were extracted with CS_2 and analyzed offline by another GC (Fig. S1, label 21, model 7890A, Agilent) equipped with an HP-5 column (HP-5, 30 m × 0.32 mm) and FID.

In each test, 3.5 ml of catalyst was diluted with clean quartz sand (sand size: 30–40 mesh) to 5 ml before loaded into the reactor for evaluation. The catalyst was *in-situ* reduced under the H_2 flow at 500 °C and 20 bar for 10 h before the FTO process. The FTO performance evaluation was carried out at a temperature of 300 °C and a total pressure of 20 bar for 80 h. The molar ratio of feed gases was H_2/CO = 2:1 and the space velocity was maintained at 1500 h^{-1}. The performance results including the CO conversions (C_{CO}), the selectivity to CO_2 (S_{CO2}), the selectivities to hydrocarbons (S_{HCs}) as well as the catalyst activity (represented by the moles of CO converted per gram catalyst per second) were subsequently calculated using the following formulas:

CO conversion (% by moles):

$$C_{CO} = \frac{M_{CO,in} - M_{CO,out}}{M_{CO,in}} \times 100\ \% = \left(1 - \frac{M_{out} \times P_{CO,out}}{M_{in} \times P_{CO,in}}\right) \times 100\%$$

CO_2 selectivity (% by moles):

$$S_{CO_2} = \frac{M_{out} \times P_{CO_2,out}}{M_{CO,in} - M_{CO,out}} \times 100\%$$

Catalysts Activity:

$$\text{Activity} = \frac{M_{CO,in} - M_{CO,out}}{m_{cata} \times t} = \frac{M_{in} \times P_{CO,in} - M_{out} \times P_{CO,out}}{m_{cata} \times t}$$

where $M_{CO,in}$ and $M_{CO,out}$, are the moles of CO at inlet and outlet; M_{in} and M_{out}, are the moles of feed and effluent gas; $P_{CO,in}$, and $P_{CO,out}$ are the mole percentages of CO at inlet and outlet, respectively; P_{C-LO} is the mole percentage of carbon that converted to light olefins. is the mole percentage of carbon that converted to light olefinsm_{cata} is the mass of loaded catalyst; t is the total reaction time.

Hydrocarbons distribution (% by weights):

$$S_{HC_s} = \frac{W_{C_n}}{W\sum C_n} \times 100\%$$

where W_{C_n} is the weight of the hydrocarbon with carbon atom number of n, $W\sum C_n$ is the total weight of hydrocarbons including the C_1-C_4 and C_{5+} hydrocarbons.

3. Results and discussion

3.1. Textural properties, structure and morphology

The compositions of the fresh catalyst samples were verified by ICP-MS and presented in Table. 1, which are consistent with the designed ratios. In addition, BET surface areas, average pore diameters, and total pore volumes of the fresh catalyst samples with different Cu loadings are also presented in Table 1. As shown in the table, due to the impregnation of Cu, the BET surface areas decrease, and the pore volumes

Table 1
Textural properties and composition of the fresh catalysts.

Catalysts	Fe/Mn/Cu/SiO_2 weight ratio from ICP-MS[a]	surface area (m^2/g)	Average pore diameter (nm)	Total pore volume (cm^3/g)
Cu-0	100/10.4/0/13.3	292.6	5.6	0.41
Cu-0.5	100/10.6/0.5/13.7	271.9	5.5	0.38
Cu-1.5	100/10.7/1.4/13.1	269.9	5.5	0.37
Cu-3.0	100/10.7/2.9/13.1	250.6	5.6	0.35

[a] Fe/Mn/Cu weight ratios were calculated based on the ICP-MS results.

of the Cu promoted samples are also slightly smaller compare with the Cu-free sample. The unchanged average pore diameter indicates that the increasing Cu content has little effect on the textural properties, which can also be verified by the BJH-pore size distribution in Fig. S2. The catalyst samples with different Cu loadings have almost identical pore size distribution. Furthermore, the average pore size of the prepared catalyst falls into the mesoporous range, which is reported to have positive influence on the FTO performance due to facile transport of intermediates formed during the catalytic reaction [21,22].

Fig. S3 shows the granule morphologies of freshly calcinated catalyst samples from microscale by SEM (S3, A1-A4) and nanoscale by TEM (S3, B1-B4), respectively. Firstly, samples with different Cu contents share a similar morphology of irregularly shaped particles with facets. The particles showing in the SEM images could consist of thousands of very small and dispersed Fe-Mn-Cu crystallites indicated by the TEM images. Thus, higher BET surface areas were obtained compared with those prepared through the same method without SiO_2 [23,24]. Secondly, with the addition of Cu, the surface from microscale looks smoother and flatter. This may indicate different types of interactions formed between the metals and SiO_2 which blocked a subset of pores on the surface. Thirdly, the TEM images in nanoscale confirm the existence of the small crystallites and also reveal that the addition of Cu doesn't affect the morphology at nanoscale. The spent catalyst samples after typical FTO processes are also shown in Fig. S3 (S3, C1–C4). It can be clearly seen that the Cu promoted samples are still in powder form, while the Cu-free sample aggregates together severely after the FTO reaction. The TEM-EDS mappings are shown in the Fig. S4, which are very similar for all the fresh catalyst samples with different Cu loadings (expect no Cu was detected on the unpromoted sample). It further implies that Cu didn't change the morphology in nanoscale. Therefore, the addition of Cu can prevent catalyst aggregation during FTO reaction by altering the catalyst surface and interactions between the metals and the SiO_2 binder, but the intrinsic structure is not affected by Cu.

The XRD patterns of freshly calcinated catalysts with different Cu loadings are presented in Fig. S5. Due to the present of amorphous SiO_2 and the interaction between metals and SiO_2, only hematite phase (α-Fe_2O_3, 2θ = 24.4°, 33.2°, 35.7°, 49.6°, 54.6°) is detected [25]. The diffraction peaks of α-Fe_2O_3 in all four samples are broad, indicating that the iron phases are well dispersed as prepared before reduction. Mn phases were not detected, probably because of the relatively low concentration and the insertion of Mn atom into the hematite lattice [26,27]. No crystallized Cu phases were detected either, due to the low concentration. The XRD patterns are very similar for the four samples before reduction.

3.2. Reduction behaviors

H_2-TPR was performed to investigate the effect of Cu on the reduction behavior of the Fe-Mn catalysts. Fig. 1A shows the H_2-TPR profiles for the catalysts with different Cu loadings. The H_2-TPR profile of the Cu-free sample is generally similar to those Cu-added samples with two main reduction regions at 300 − 500 °C (Region I) and 600 − 850 °C (Region II). The shaded peaks under the H_2-TPR curves are simulated to generally represent the reduction state. The peaks in Region I at 300 − 500 °C are assigned to the reductions of CuO, α-Mn_2O_3, and α-Fe_2O_3 to Cu, MnO, and FeO/Fe_3O_4, respectively. The peaks in Region II at higher temperature (600 − 850 °C) are corresponded to the further reduction of FeO/Fe_3O_4 to Fe [28].

It is shown that all the iron oxides in the four catalysts can be finally reduced to Fe in the H_2- TPR process at a temperature up to 1000 °C. In addition, there are some noticeable differences in the H_2-TPR profiles of these catalysts. Firstly, there is a trend that the reduction peaks generally shift toward lower temperature with the addition and increasing amount of Cu promoter, indicating an obvious facilitating effect of Cu on the reduction process. Secondly, the low temperature reduction peak in Region I separates into two small peaks and shifts to lower temperature with the loading amount of Cu from 0 to 1.5 wt%. Those peaks overlap into a single large peak on the sample with 3.0 wt% Cu added. This implies that the addition of Cu may change the interaction between metal oxides and SiO_2 to some extent, and significantly lowers the required reduction temperature. Thirdly, in the Region II reduction, the first reduction stage is facilitated with the increasing Cu content, which is evidenced by the shifting reduction peak. The second reduction stage in Region II, representing the reduction to metallic Fe, is promoted by the addition of Cu, but doesn't change much by further increasing the Cu content. Therefore, the addition of Cu facilitates the reducibility of the metals probably by changing the interactions between the metal oxides and SiO_2. However, further increasing the amount of Cu doesn't change the final step of reduction to metallic Fe.

The H_2-TPR profiles imply the changing of interactions between metals oxides and silica to some extent. The H_2-DTG results, shown in Fig. 1B, further confirmed these kinds of interactions. The first runs of H_2-DTG are consistent with the H_2-TPR results, which show two main reduction regions at 300 − 500 °C and 600 − 850 °C. The maximum reduction rates at the temperature below 500 °C increases as the Cu content increases from 0 to 3.0 wt%, verifying that Cu could improve the reduction efficiency. For the reduction at higher temperature, the addition of Cu improves the reduction rate; however, the further increasing of Cu content has limited influence on the reduction efficiency, which is also consistent with the results from H_2-TPR.

After the first run of H_2-DTG, all samples were oxidized under oxygen before the second run. Compared with the first run profiles of all the four samples, the onsets of the reduction process are delayed in the second run and the reduction rates at low temperature are obviously lower than the first run. Meanwhile, the temperatures at which the reduction rate is maximum are also delayed. The delays of the reduction onsets, as well as the decreased reduction rates in the second run of H_2-DTG tests are due to the aggregation of metal oxides during the first run and re-oxidation process, which makes the reduction process harder. For the low temperature reduction in the second run, no reduction process is observed for the Cu-free sample until the temperature reaches 500 °C. As for the Cu-added samples, lower incipient reduction temperatures are observed compared with the Cu-free sample, which means the addition of Cu can still facilitate the reduction process even as aggregation is present. For the high-temperature reduction in the second run, the profiles for Cu-free and Cu-promoted samples are almost the same, the maximum reduction rate peaks are at the same temperature and smaller than those in the first run. This result indicates the loss of facilitation ability of Cu on this stage of the reduction process.

Zhang et al. reported that copper in the precipitated catalyst samples have interactions between metal oxides and SiO_2 at the same time [18]. The interaction of copper with metal oxides can facilitate the reduction process at low temperature by providing H_2 dissociation sites when CuO is reduced to Cu [29]. Meanwhile, the interaction of copper with SiO_2 can ease the reduction at higher temperature by further interacting with metal oxides that haven't been fully reduced during the low temperature reduction. In this case, during the oxidation process before the second run reduction, all reduced metals were re-oxidized and the interactions between metals and SiO_2 were destroyed. The CuO on the surface can still interact with Fe and Mn and promote their first stage reductions. However, the lost interactions between metals and SiO_2 makes the high temperature reduction behavior the same for both Cu-free and Cu-added samples.

3.3. Surface basicity

CO_2-TPD was performed to investigate the effect of Cu on the surface basicity of the catalysts. The profiles are presented in Fig. 2. The CO_2-TPD profiles for the fresh samples with different Cu loadings are almost identical in Fig. 2 (the grey curves), while obvious differences showed up after the reduction process. It suggests that Cu promoter affected the surface basicity during the reduction process, which is

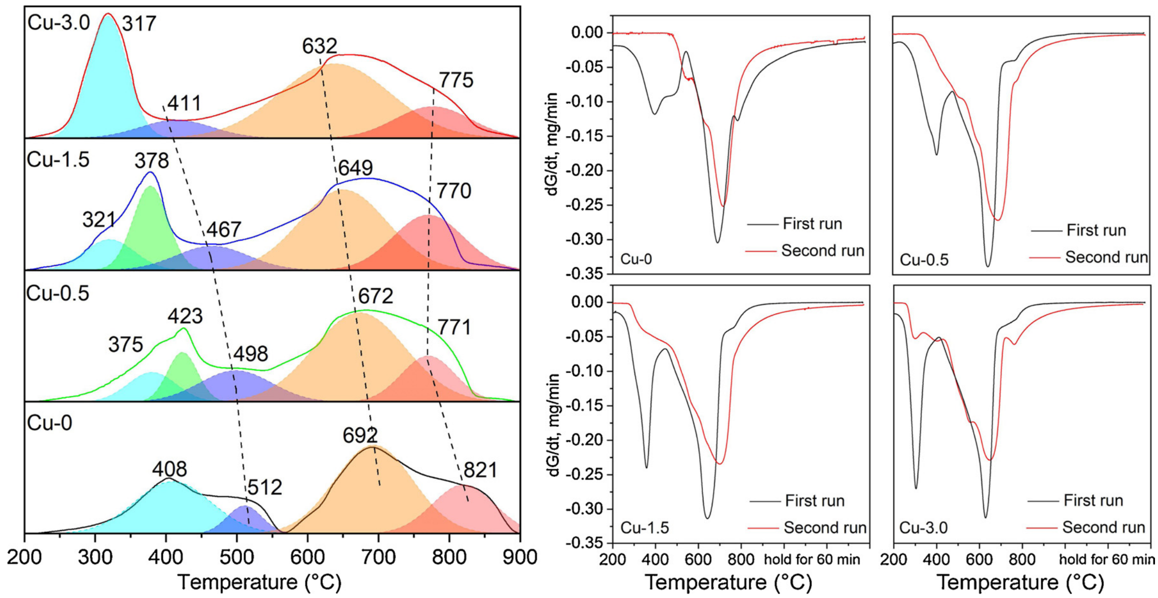

Fig. 1. (A) H_2-TPR and (B) H_2-DTG profiles for Catalyst with different Cu loadings (Fist run: Reduction of fresh catalyst; Second run: Reduction of the *in-situ* oxidized catalyst after the first run).

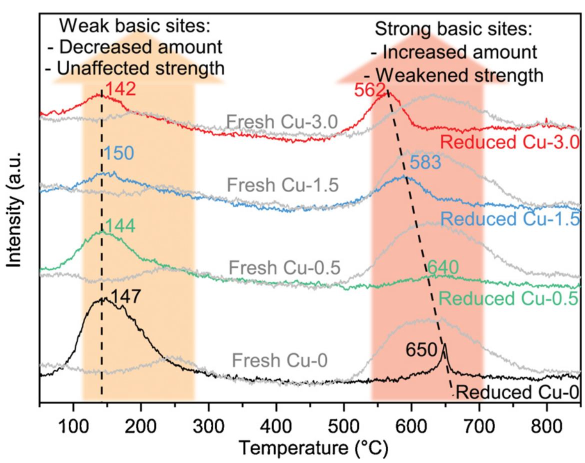

Fig. 2. CO_2-TPD profiles of the fresh and reduced catalyst samples with different Cu loading amounts (reduced samples were pre-treated at 500 °C for 2 h under H_2 flow prior to the TPD tests).

consistent with the different reduction behaviors revealed by the H_2-TPR and H_2-DTG tests in Fig. 1. For the reduced catalyst samples in Fig. 2, it clearly shows that all samples have two groups of desorption peaks, indicating two different kinds of basic sites on the surface of the catalysts. The desorption region at low temperature (150 – 300 °C) corresponding to the desorption of CO_2 from weak basic sites, and the peaks at high temperature (550 – 700 °C) are ascribed to the desorption of chemical adsorbed CO_2 at strong basic sites. The amount of desorbed CO_2 and the temperature at which the CO_2 desorption is at maximum indicate the amount and the strength of basic sites, respectively [30,31]. The quantitative results of the CO_2 uptake for the reduced catalyst samples in the CO_2-TPD are presented in Table S1, which are determined by integrating the corresponding peak area for each CO_2-TPD curve and compared with the results of calibration gas with known amount passed through the TCD. For the weak basic sites, with the increasing Cu content, the desorption peak areas clearly decrease but the positions are similar, suggesting that the addition of Cu results in decrement of weak basic sites but doesn't change the strength of the adsorption on those basicity sites. For the desorption of CO_2 at high temperature, the peaks shift toward the lower temperature with increasing Cu content, which reveals a negative effect of Cu on the strong surface basicity. However, the amount of the strong basicity for Cu promoted samples are increasing gradually with the increasing Cu content, suggesting that the addition of Cu results in an increment of strong basicity sites even though the strength of those sites is weakened. This result may due to the formation of Fe_2SiO_4 as well as other more reduced iron species during the H_2 reduction process according to the XRD and H_2-TPR results. It is noted that the high temperature desorption peak for the Cu-free sample is relatively sharp, while the Cu-promoted samples have broader peaks. This implies that different types of strong basicity sites are formed in the presence of Cu. Furthermore, the total CO_2 uptake as presented in Table S1 significantly decreased from 67.06 μmol CO_2/g to 36.22 μmol CO_2/g when 0.5 wt% Cu was added but kept almost unchanged when more Cu content was added. This is due to the promotion effect of Cu on the reducibility of iron oxide. As presented in H_2-TPR (Fig. 1A), the high temperature reduction that represents the formation of metallic Fe was improved by the addition of Cu but didn't change much when further increasing the Cu content. In brief, the decreased weak CO_2 adsorption and increased strong CO_2 adsorption suggests that the increasing Cu loadings promoted the formation of more strong basic sites during the reduction process. The total CO_2 uptake for Cu-promoted samples were almost unaffected, manifesting that the increasing Cu content didn't affect the total amount of basic sites. Meanwhile, the promoter Cu also weakened the strength of strong basic sites but no obvious effect on the strength of weak basic sites.

Decreased surface basicity has a negative impact on chain growth during a FT process and could result in an increased CH_4 selectivity and fewer long-chain hydrocarbons [32,33]. Additionally, the stable and strong CO_2 adsorption may lead to high partial pressure of CO_2 on the surface and suppress the water gas shift (WGS) reaction.

3.4. Chemisorption behavior

The CO and H_2 adsorption behavior are illustrated by the H_2/CO-TPD profiles (Fig. 3). Fig. 3A shows the CO desorption behavior on the surface of the catalysts. Two main stages of CO desorption for all catalysts can be recognized. The desorption at low temperature can be assigned to the weak adsorption of CO on metal oxides, while the high temperature desorption is from the dissociate adsorption of CO on the surface [34]. For the low temperature desorption, the three desorption

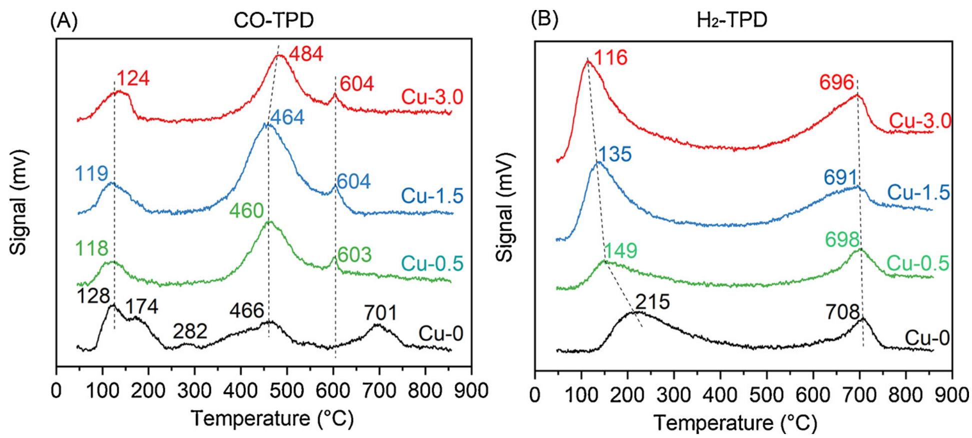

Fig. 3. H_2/CO-TPD profiles of the reduced catalysts with different Cu loading amounts (all samples were *in-situ* reduced at 500 °C and H_2 flow for 2 h prior to the TPD test).

peaks below 300 °C for Cu-0 sample suggest a variety of iron oxide adsorption sites due to lower reduction degree. Contrastingly, in the presence of Cu, only one desorption peak is apparent at low temperature. For the high temperature desorption, clear differences can be distinguished between the Cu-free and Cu-promoted samples. The peak at 701 °C for the Cu-0 sample shifts to lower temperature and merged into the peak at 466 °C to some extent, which implies different types of adsorption sites formed in the present of Cu. It is also consistent with the CO_2-TPD results in Fig. 2. The peak areas are increased with addition of Cu, pointing to an overall enhanced CO dissociative adsorption. Furthermore, for both low and high temperature desorption, with further increasing of Cu, no significant difference is observed, suggesting that 0.5 wt% of Cu is sufficient to alter the surface CO adsorption behavior.

In Fig. 3B, two H_2 desorption regions reveal the existence of two different types of H_2 adsorption on the catalyst. The peaks at $100-300$ °C are ascribed to the chemisorbed H_2 on the metal surface, while the peaks at higher temperature (around 700 °C) describe the dissociated H_2 on the surface. The catalyst samples with different Cu loadings have very similar peaks at high temperature desorption, indicating the addition of Cu doesn't affect the dissociative adsorption of H_2. However, the desorption peaks at low temperature for the four samples are quite different. The positions of peaks shift to lower temperature while the peak areas increase with the addition and increasing amount of Cu. This trend implies that the added Cu promoter interacts with other metals and SiO_2 and generates more weaker H_2 adsorption sites during the reduction process.

Fig. 4 shows the FTIR profiles for the samples with different Cu loadings. Fig. 4A–D show the development of the *in-situ* FTIR spectra, in which the samples were heated *in-situ* from 30 to 300 °C at 10 °C/min under pressurized condition with CO and H_2. Fig. 4E and 4F are the spectra of the samples at T = 15 min (180 °C) and T = 30 min (300 °C), respectively. The broad peaks in the region of $2250-2050$ cm^{-1} correspond to the gas CO, and the bands at 2032 cm^{-1} and 2014 cm^{-1} are attributed to the CO adsorption on the metallic Fe [35–37]. The bands at 2367 cm^{-1} and 2337 cm^{-1} are assigned to the P and R branches of CO_2 in gas phase, which can be considered as the sign of an active FT reaction [38]. Very similar IR spectra are observed for the samples with different Cu loadings, however, there are slight differences for the CO adsorption at bands of 2032 cm^{-1} and 2014 cm^{-1}. It should be noted that the bands of CO_2 appear when the bands of CO adsorption on Fe° begin to decrease. The decreased intensity of bands at 2032 cm^{-1} and 2014 cm^{-1} was due to the phase transformation of metallic iron to iron carbide, which is the active phase for the FT reaction (as identified by XRD in Fig. 6). The bands of CO_2 in gas appeared once the active phase formed and the reaction began to happen. In this case, the bands of CO adsorption on Fe° disappeared when all the metallic iron was carbonized. The duration of CO bonding on Fe° at 2032 cm^{-1} and 2014 cm^{-1} for the Cu-free sample was longer compared with the Cu-promoted ones, manifesting that the carbonization process was more efficient in the presence of Cu. Noting H_2-TPR suggests harder reductions of the metal oxides when Cu is not present (Fig. 1A), the more efficient carbonization process are likely resulted from the better reduction of the catalyst. In addition, the delayed appearance of CO bonding on Fe° for the Cu promoted samples may also indicate more adsorped H_2 at the beginning of the reaction since H_2 and CO have a competitive adsorption relationship.

3.5. *FTO performance*

3.5.1. *Conversion, activity, and CO_2 selectivity*

The FTO performances of the catalysts with different Cu loadings were evaluated for 80 h in a fixed bed reactor under the condition of 300 °C, 20 bar, H_2/CO = 2.0 and space velocity of 1500 h^{-1}. To better demonstrate and understand the results from this study, main results from some literatures are also listed and labeled as Ref. 1, Ref. 2 and Ref. 3 respectively [1–3]. More works from literatures were compared in Table S2.

The CO conversions for samples with different Cu loadings are shown in Fig. 5A. The overall CO conversion was increased from 74 % to around 95 % with the addition of Cu. It took about 10 h to achieve maximum CO conversion on all samples, but the catalysts with addition of Cu maintained stable at conversions of 95 % beyond 15 h, whereas the sample without Cu began to decrease. The higher CO conversion when Cu promoter was added can be explained by a combination of improved reduction and enhanced CO dissociative adsorption, as evidenced by H_2-TPR (Fig. 1A) and CO-TPD (Fig. 3A). These improvements lead to a better carburization process to provide more active sites for the FTO reaction.

Combining these reaction results and CO_2-TPD results suggests that catalyst deactivation is inhibited due to the overall weakened basicity by the addition of Cu, which has negative effect on carbon deposition. [39] The inhibition of the catalyst aggregation with addition of Cu, as evidenced by the spent catalyst maintaining its powder form (Fig. S3) also contribute to the improved activity. Thus, the Cu promoter improves catalyst activity and stability in the Fe-Mn-Cu-SiO_2 system.

Fig. 5B shows the activities (represented by the moles of CO converted per gram of catalyst per second) and the CO_2 selectivities (based on the mole percentage of converted CO). Compared with some of the best-reported FTO results, at a similar catalyst activity level, a CO_2 selectivity as low as 23 % was achieved in this study, which is significantly lower than most of the reported results [1–3,7,8,40,41]. In other words, as high as 77 % carbon conversion efficiency was achieved for all catalyst samples in this study. CO_2 is generated through the WGS

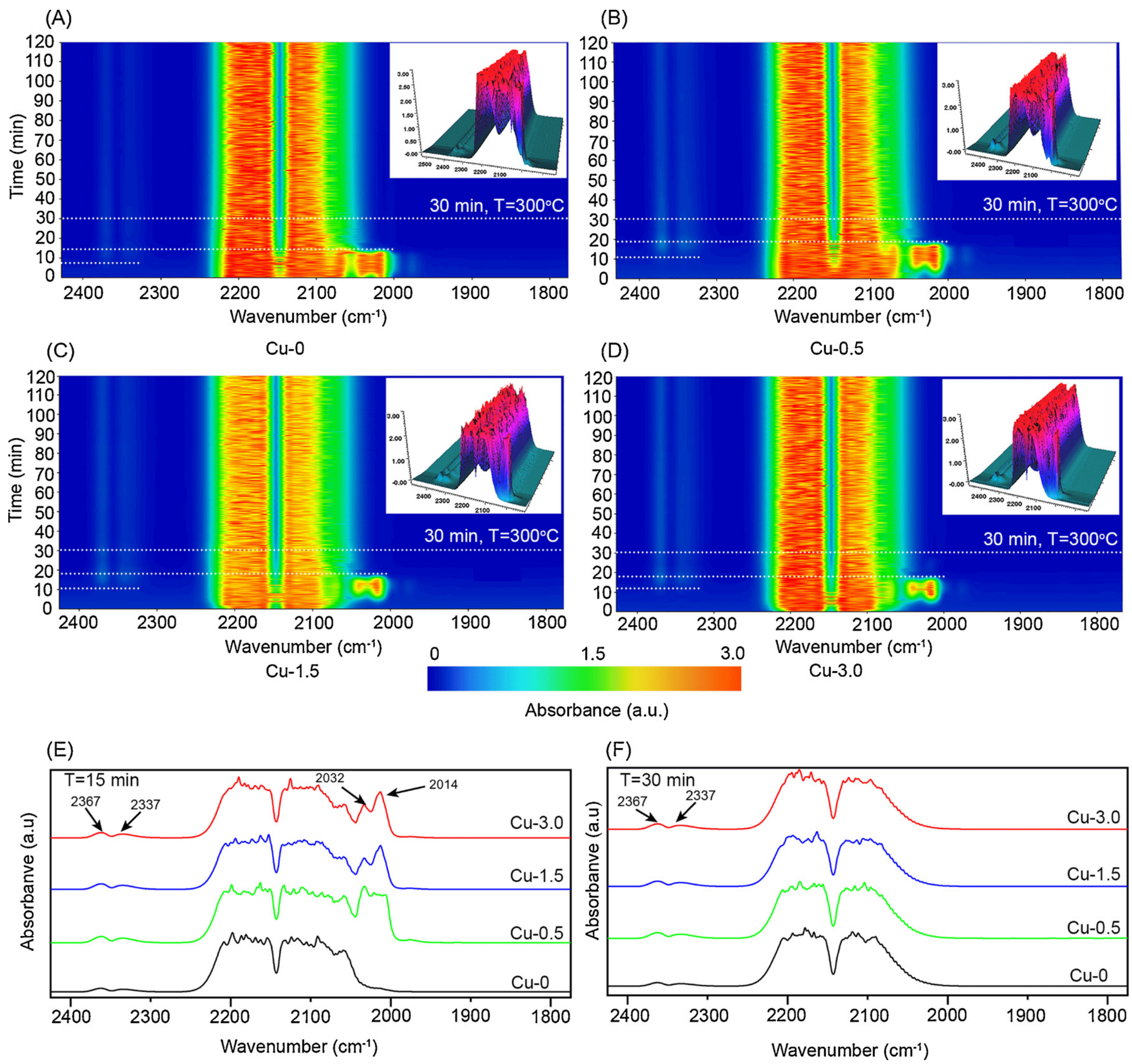

Fig. 4. FTIR spectra for samples with different Cu loadings. (A-D: *in-situ* FTIR spectra under CO and H_2 from 30 to 300 °C at 10 °C/min; E: FTIR spectra at T = 15 min; F: FTIR spectra at T = 30 min).

reaction during the FTO process, and a lower CO_2 emission means a suppressed WGS reaction. Since WGS reaction is a moderately exothermic reversible process with $\Delta H^{\circ}_{298} = -41$ kJ/mol, it is thermodynamically favored but kinetically unfavored at the relatively low temperature (300 °C) [42]. As Teng et al. and Bukur et al. reported, the WGS reaction under FTS reaction conditions is far from equilibrium, and the increased temperature will enhance the WGS reaction mainly due to kinetic effects, which contributes to the lower CO_2 emission in the current FTO tests [43,44]. Another reason for the low CO_2 selectivity is the strong and stable CO_2 adsorption, as evidenced by the existence of desorption peaks at high temperature for all samples in the CO_2-TPD tests. The stable adsorption could result in higher CO_2 coverage and increased partial pressure on the surface during the reaction, which hindered the WGS reaction. In addition, the feeding gas ratio of H_2/CO is around 2.0, and the high H_2 ratio and adsorbed H_2 coverage on the surface could also suppress the WGS reaction.

3.5.2. Product distribution

The hydrocarbon product distribution that is normalized with exception of CO_2 is shown in Table 2 and Fig. 5C. The selectivity to light olefins increases from 31.3%–40.1% with the Cu loading increases from 0 to 3.0 wt%. The selectivities of CH_4 and light paraffins increase simultaneously while the C_{5+} decreases from 32.7 %–9.5 %. It implies that the addition of Cu increased the yield of light olefins by shifting the product distribution towards lower molecular weight products in proportion. In other words, the addition of Cu lowered the chain growth probability and favored the formation of short-chain hydrocarbon products. It can be rationalized by the chemisorption behavior and the surface basicity properties. The CO-TPD profiles indicate an overall enhanced CO dissociative adsorption and the H_2-TPD shows an increased amount of adsorbed H_2 with the addition of Cu promoter. The enhanced CO dissociative adsorption and increased H_2 adsorption amount favored the formation of CHx intermediates during the FTO reaction. However, the weakened strength of the strong basicity sites as evidenced by the CO_2-TPD (Fig. 2) hindered the adsorption of the formed CHx intermediates on the surface, which diminished the chain propagation ability on the surface during the reaction and resulted in increasing selectivity to light hydrocarbons.

A typical FT process is a surface-catalyzed stepwise polymerization reaction and the chain termination can be achieved by either hydrogenation to form paraffins or by H elimination to form olefins. The unaffected H dissociative adsorption indicated by H_2-TPD results and the faster desorption of CHx intermediates due to weakened surface basicity will cause increased coverage of H* on the surface, which favors chain termination by hydrogenation. However, the CO_2-TPD also reveals the incremental amount of strong surface basicity sites. As a

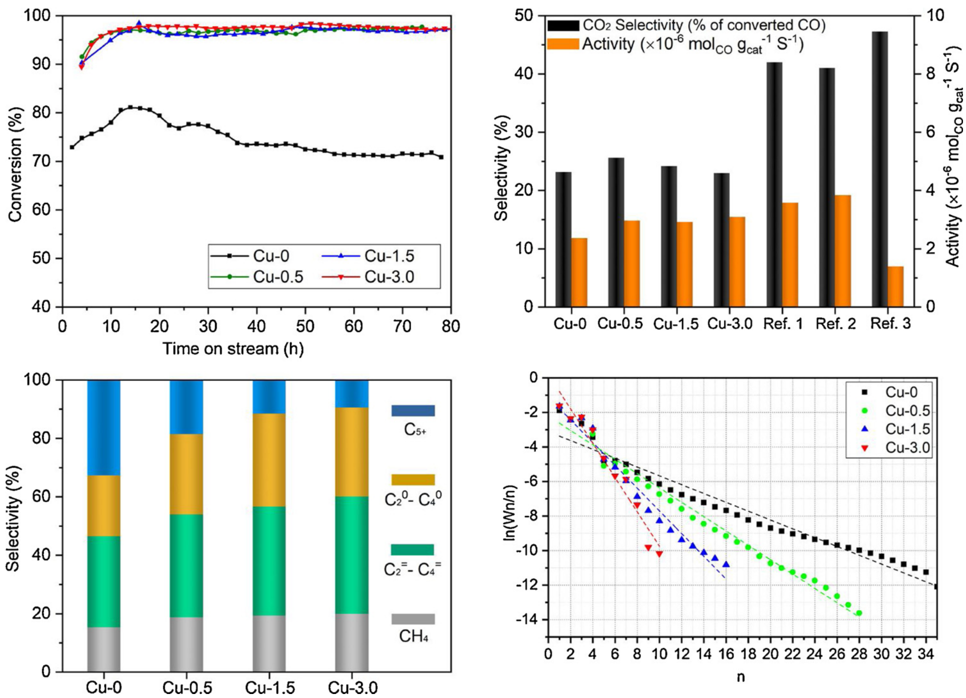

Fig. 5. (A) CO conversion of catalyst samples with different Cu loadings; (B) Comparisons of prepared catalysts with Ref. [1] (340 °C, 20 bar, $H_2/CO = 1$, Torres Galvis et al., 2012, Science [1]), Ref. 2 (400 °C, 25 bar, $H_2/CO = 2.5$, Jiao et al, 2016, Science [2]) and Ref. 3 (250 °C, 1 bar, $H_2/CO = 2$, Zhong et al., 2013, Nature [3]) in terms of CO_2 selectivity and activity; (C) Selectivities to hydrocarbons; (D) ASF distribution plots.

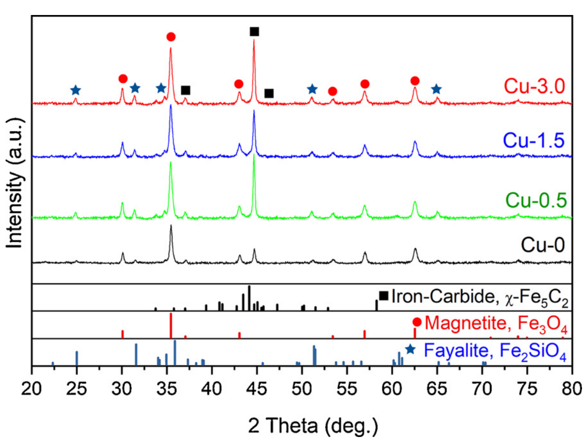

Fig. 6. XRD patterns of spent catalysts with different Cu loadings.

result, the increasing trend of H* species coverage on the surface is suppressed due to the competitive adsorption with more CHx intermediates; therefore, the possibilities of hydrogenation and H elimination for chain termination doesn't change much with the addition and increasing amount of Cu content. This explains that the olefin/paraffin ratios don't change significantly with the increasing amounts of Cu. Generally, the product distribution can be described and predicted by the Anderson-Schulz-Flory (ASF) theory with a chain growth probability variable (α). As shown in the Fig. 5D, with an increase in the Cu loading, the α values increase from 0.77 to 0.37, which is near the theoretically optimized value (α≈0.4) for the production of light hydrocarbons [45].

3.5.3. Active phase in FTO

Fig. 6 shows XRD patterns of the spent catalysts. The catalyst samples after reaction are significantly more crystalline compared with the as-prepared samples shown in Fig. S5. Hägg iron carbide (χ-Fe_5C_2, ICDD PDF# 36-1248, 2θ = 40.9°, 43.5°, 44.2° and 45.1°), magnetite (Fe_3O_4, ICDD PDF#74-0748, 2θ = 30.1°, 35.4°, 37.1°, 43.1°, 56.9°, and 62.5°), and fayalite (Fe_2SiO_4, ICDD PDF# 34-0178, 2θ = 25.0°, 31.6°, 34.9°, 35.9° and 51.3°) are identified for all four samples. No iron carbide other than χ-Fe_5C_2 is detected by XRD, which agrees with

Table 2
FTO performance of catalysts.

Samples	CO conversion (%)	CO_2 selectivity[e] (%)	Activity 10^{-6} mol_{CO} g_{cat}^{-1} s^{-1}	Hydrocarbon product distribution[f]			
				CH_4	$C_2^{=}$-$C_4^{=}$	C_2°-C_4°	C_{5+}
Cu-0[a]	74.8	23.2	2.37	15.3	31.3	20.8	32.7
Cu-0.5[a]	96.5	25.6	2.97	18.8	35.2	27.5	18.5
Cu-1.5[a]	93.1	24.2	2.92	19.4	37.2	31.7	11.6
Cu-3.0[a]	96.9	23.0	3.09	20.0	40.1	30.4	9.5
Ref. 1[b] [1]	88.0	42.0	3.58	13.0	52.0	12.0	18.0
Ref. 2[c] [2]	17.0	41.0	3.84	2.0	80.0	14.0	4.0
Ref. 3[d] [3]	31.8	47.3	1.39	5.0	60.8	2.0	31.4

[a] Reaction conditions: T = 300 °C, P = 20 bar, GHSV = 1500 h^{-1}, $H_2/CO = 2$, mass balance < 5 % error.
[b] Reaction conditions: T = 340 °C, P = 20 bar, GHSV = 1500 h^{-1}, $H_2/CO = 1$.
[c] Reaction conditions: T = 400 °C, P = 25 bar, GHSV = 5143 ml/h·g_{cat}, $H_2/CO = 2.5$.
[d] Reaction conditions: T = 250 °C, P = 1 bar, GHSV = 2000 ml/h·g_{cat}, $H_2/CO = 2$.
[e] based on the moles of converted CO.
[f] weight percentage, CO_2 free.

previous literature suggesting that χ-Fe_5C_2 is the active sites for FTS at around 300 °C [46,47]. In this work, as well as the works done by de Jong, Cheng and their co-workers confirm that the active phase for FTO over Fe-based catalyst is still χ-Fe_5C_2, which is responsible for initializing the chain growth reaction [7,48]. In addition, the diffraction peaks of χ-Fe_5C_2 are intensified for Cu-added samples, pointing to a promoting effect of Cu on the formation of active phases during the reaction. Fe_2SiO_4 that is formed by the intimate contact between iron and silica confirms the Fe-SiO_2 interaction [25,49]. The Fe_2SiO_4 phase becomes more crystalline with increasing Cu content, suggesting that the addition of Cu improves the interaction between Fe and the binder SiO_2, which in turn favors the formation of χ-Fe_5C_2 when in contact with syngas at the beginning of the reaction. Considering the promotional effect of Cu on the reduction process, the improved Fe-SiO_2 interaction may form during the reduction process, where an alternative reduction pathway may be provided by addition of Cu [50].

3.5.4. Mechanism

To better understand the effect of Cu on the production of light olefins *via* FT process, a discussion of these results in context of the generally accepted alkyl mechanism and H-assisted CO dissociation is provided (Fig. 7) [51,52]. Ojeda, et al. reported that CO dissociates on the catalyst surface through a combination of two parallel routes: direct and H-assisted dissociation. The H-assisted activation route is kinetically and thermodynamically favored with a lower forward activation energy barrier (E_f) of 89 kJ mol^{-1} compared with 189 kJ mol^{-1} for the unassisted direct CO dissociation [53]. In direct CO dissociation, CO* dissociates to C* and O*, and C* subsequently reacts with H* to form CHx* monomers. O* is either removed by CO_2 *via* reaction with CO* or by H_2O *via* reaction with H*, the first one is more preferential. In the H-assisted CO dissociation, CO* forms HCO* and then HCOH* *via* reaction with H*, followed by HCOH* dissociation to CH* that ultimately forms the initiator and monomer for chain growth. OH* groups are removed by H_2O. Both routes can form the CHx* intermediate species that contribute to the chain growth reaction. The hydrocarbon chain grows by the addition of methylene (CH_2*) into the adsorbed alkyl species and the coverage rate of the intermediates is the primary influence on chain propagation behavior. The polymerization is terminated by either β-H elimination to form olefins or by hydrogenation to produce paraffins. As highlighted in the graph, the addition of Cu diminished the adsorption strength of C* and CHx* intermediates, enhanced the chemisorption of H_2, which favored the H-assisted route for CO dissociation. While CHx* increased on the surface, its faster desorption leads to a lower chain growth probability and the product shifting to lower hydrocarbons. In addition, the promoted H-assisted CO dissociation route increases the CO dissociation rate, which results in improved activity, indicated by the higher CO conversion in the presence of Cu. In the meantime, more formed CHx* intermediates offset the trend of hydrogenation termination of the chain growth reaction brought on by the higher H* coverage on the surface. Therefore, the olefin-to-paraffin ratio was not significantly affected with the addition of Cu.

4. Conclusions

Fe-Mn based catalyst samples with different concentrations of Cu promoter were prepared and evaluated for FTO performance. Several characterization techniques were combined to synergistically investigate the effect of Cu on the textural properties, reduction process, surface basicity, chemisorption behavior, as well as FTO performance. The Cu promoter was found to facilitate the reduction process and enhance CO dissociative adsorption by changing the interactions between Fe, Mn and the SiO_2 binder. The resultant catalytic activities and overall CO conversions increased from 2.37 $\mu mol_{CO}\ g_{cat}{}^{-1}s^{-1}$ and 74.8 % to 3.09 $\mu mol_{CO}\ g_{cat}{}^{-1}\ s^{-1}$ and 96.9 %, respectively. More stable catalytic activities were achieved on Cu-promoted samples due to reduced carbon deposition, resulting from the weakened surface basicity. The addition of Cu lowered the chain growth probability and achieved 40.1 % selectivity of light olefins by favoring H-assisted CO dissociation and shifting the hydrocarbon product toward short-chain hydrocarbons proportionally. CO_2 selectivity as low as 23.0 % was obtained due to the suppressed WGS reaction, which resulted from the strong and stable CO_2 adsorption during the reaction. The obtained results provide a

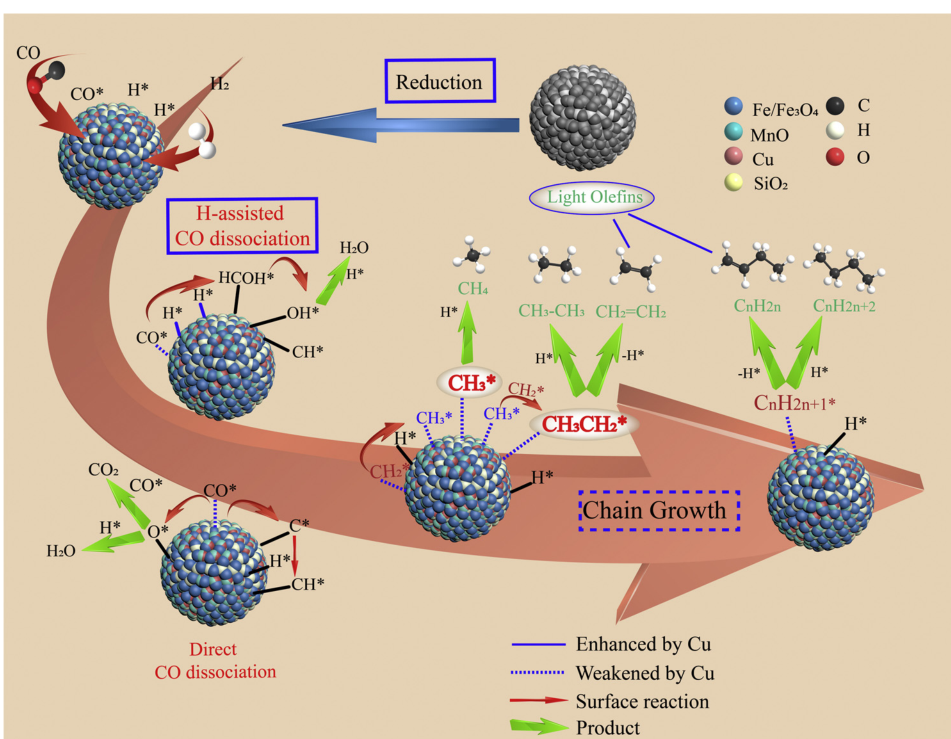

Fig. 7. A proposed schematic reaction network for the production of light olefins *via* FT process.

fundamental guidance for modification of the conventional FT catalyst for production of light olefins with relatively high activity and carbon efficiency.

CRediT authorship contribution statement

Weibo Gong: Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft, Writing - review & editing. **Runping Ye:** Conceptualization, Methodology, Validation. **Jie Ding:** Methodology, Formal analysis. **Tongtong Wang:** Visualization, Validation. **Xiufeng Shi:** Writing - review & editing. **Christopher K. Russell:** Writing - review & editing. **Jinke Tang:** Resources. **Eric G. Eddings:** Writing - review & editing. **Yulong Zhang:** Supervision. **Maohong Fan:** Project administration, Resources, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Supplementary material

Effect of Copper on Highly Effective Fe-Mn Based Catalysts During Production of Light Olefins via Fischer-Tropsch Process with Low CO_2 Emission

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Table of Contents

Materials

Iron nitrate (III) nonahydrate ($Fe(NO_3)_3 \cdot 9H_2O$, Sigma-Aldrich ACS reagent, >98%), Manganese (II) nitrate tetrahydrate ($Mn(NO_3)_2 \cdot 4H_2O$, purum p.a., >97.0%, KT) and Copper (II) nitrate hemi(pentahydrate) ($Cu(NO_3)_2 \cdot 2.5H_2O$, Sigma-Aldrich ACS reagent, >98%) were used to provide the appropriate amount metal loading in the catalyst. Silica gel (Sigma-Aldrich high-purity grade, pore size 6 nm, 200-400 mesh particle size) was used as a binder material in the catalyst. Ammonium hydroxide (NH_4OH, Fisher Chemical, Certified ACS Plus, 28-30%) and Nitric acid (HNO_3, Sigma-Aldrich, ACS reagent, 70%) were used to prepare the appropriate precipitation environment during the preparation process.

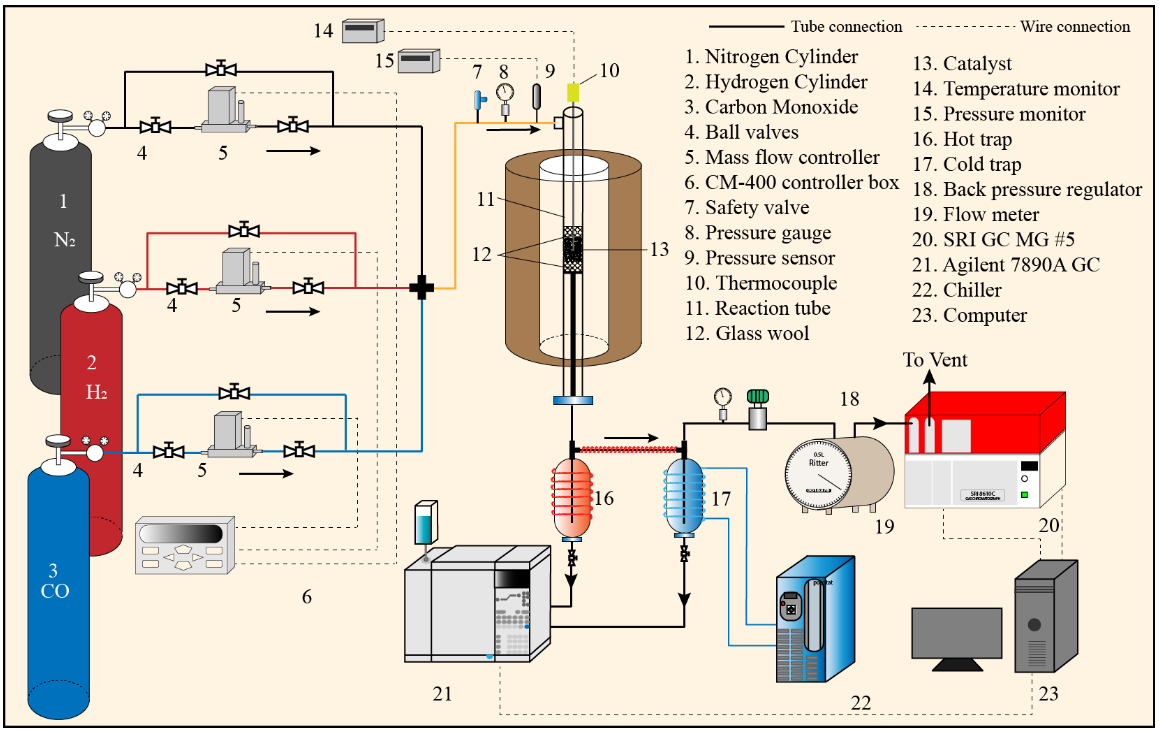

Figure S1. The schematic diagram of the FTO examination system

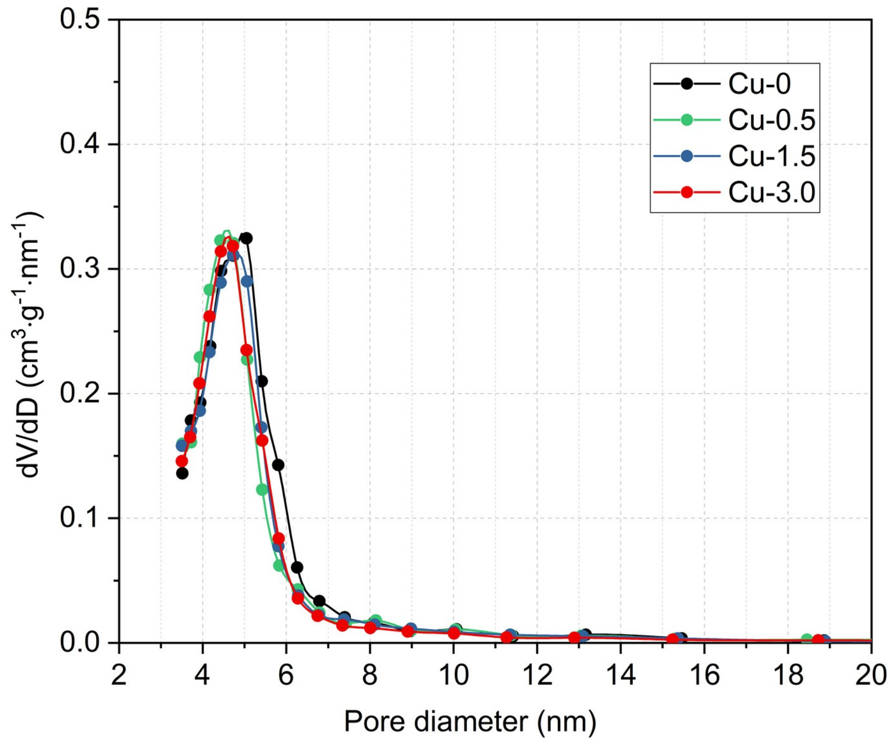

Figure S2. BJH-pore size distribution of catalysts with different Cu loadings

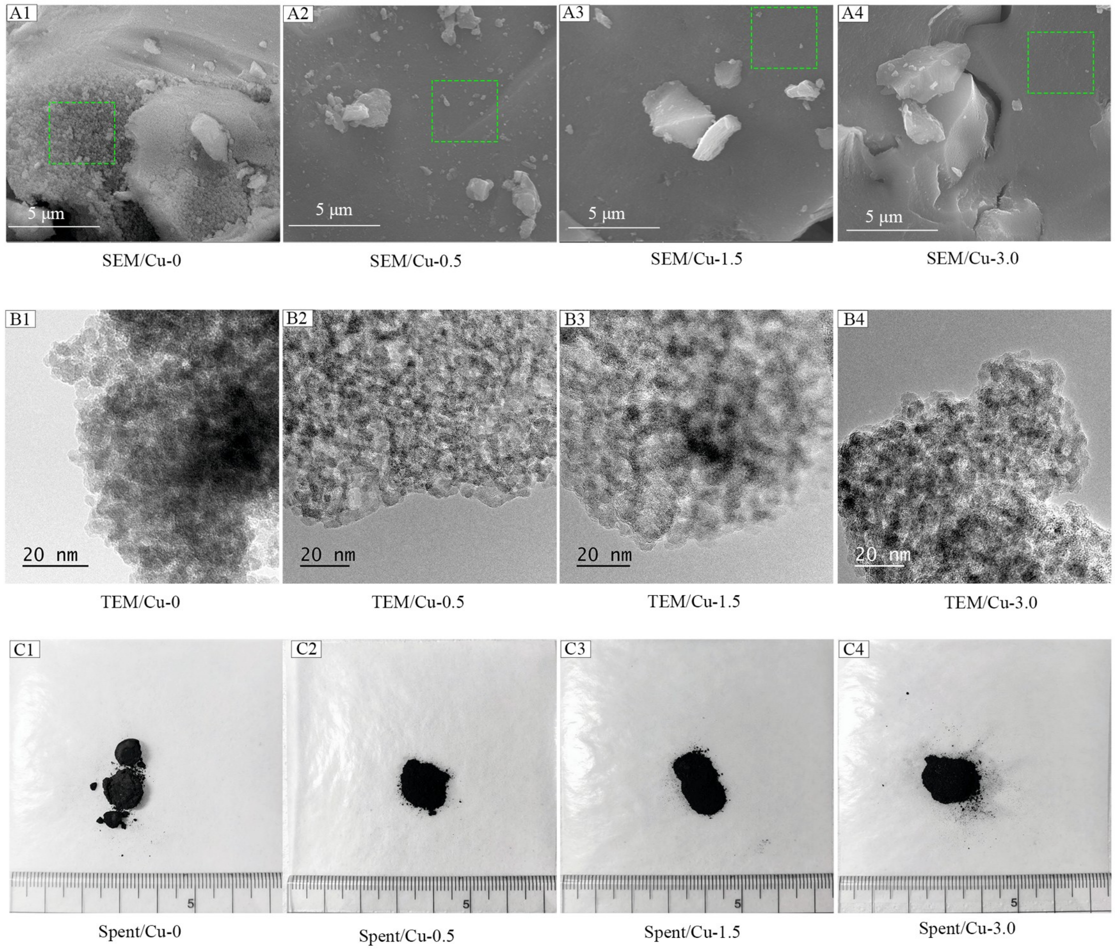

Figure S3. Morphology of catalysts with different Cu loadings. A1-A4: SEM images, B1-B4: TEM images, C1-C4: Spent catalyst

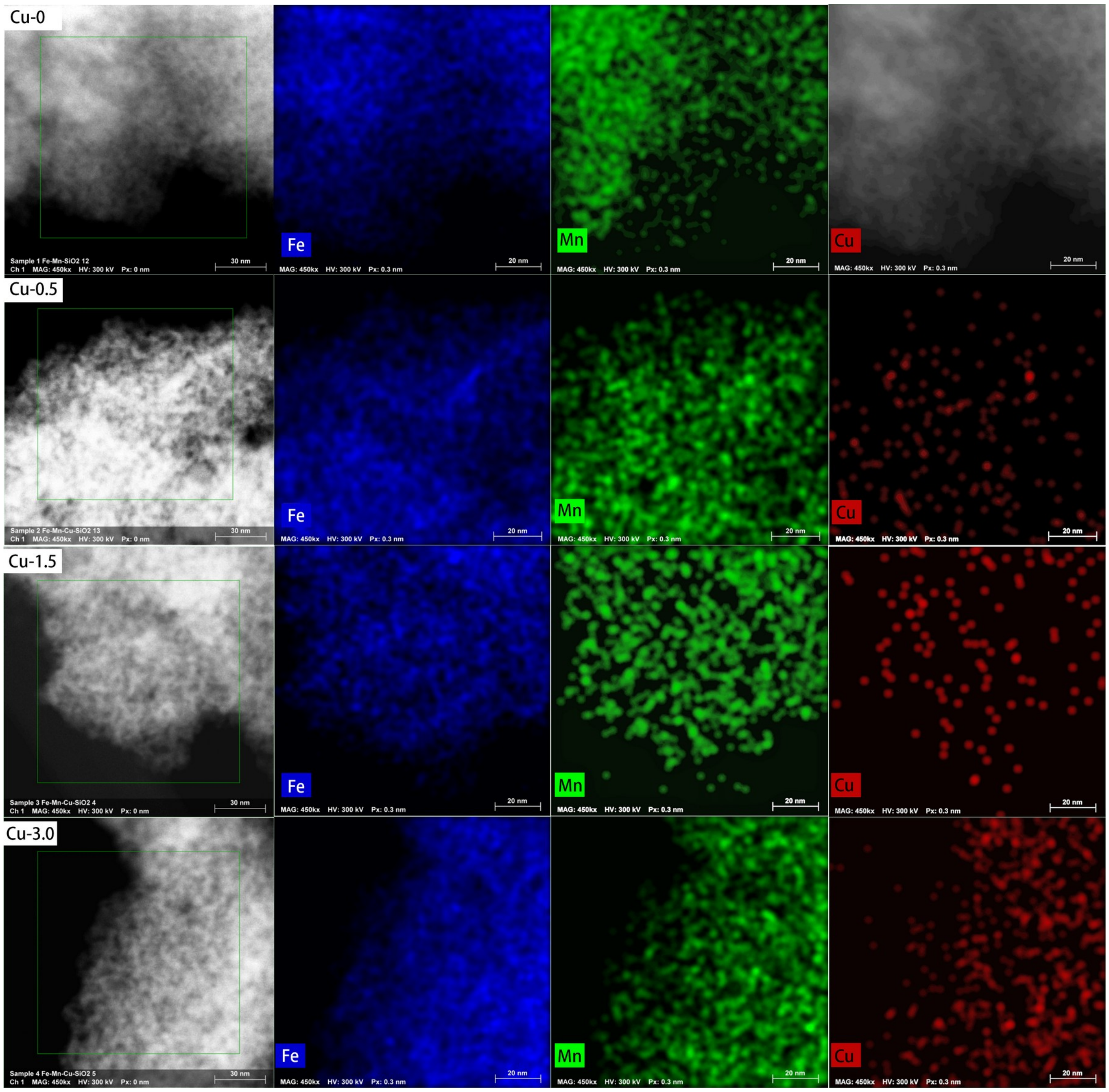

Figure S4. The TEM-EDX images of catalysts with different Cu loadings

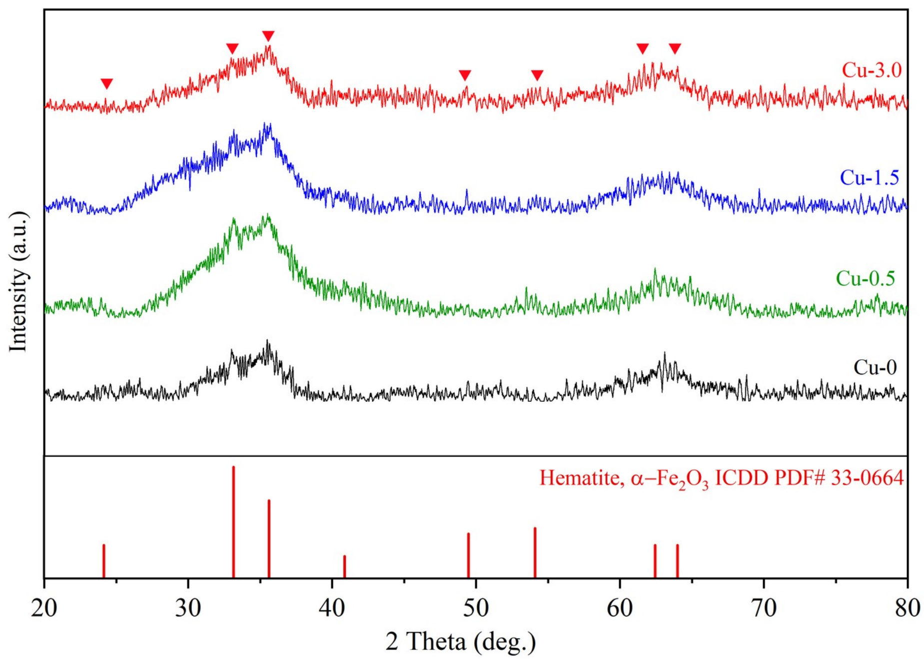

Figure S5. XRD patterns of as-prepared catalyst with different Cu loadings

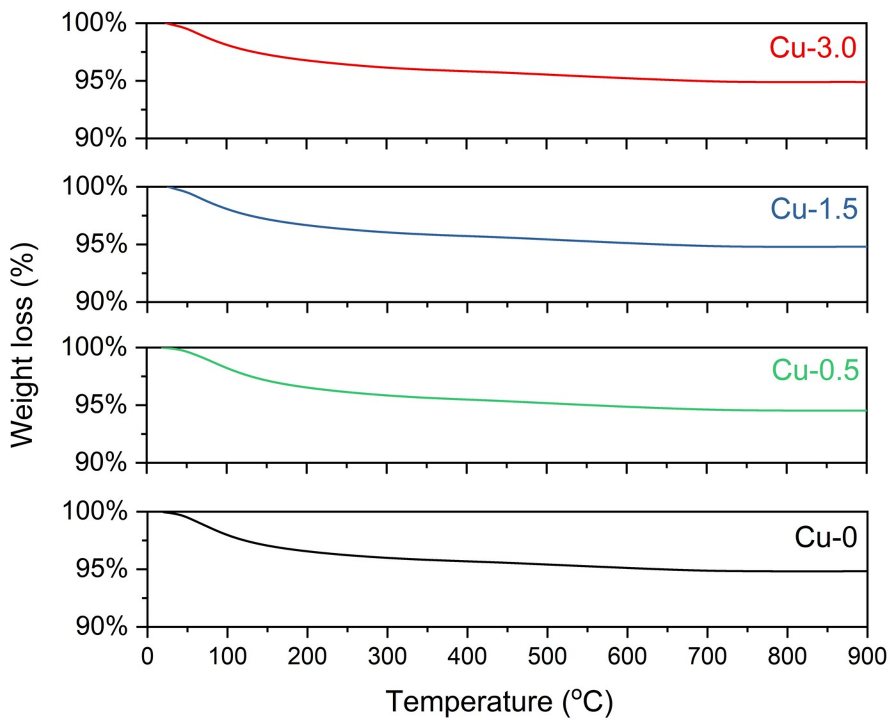

Figure S6 TGA profiles of the fresh catalysts with different Cu loadings (50 mg of catalyst, 10 °C/min to 900 °C)

Table S1. Quantitative results of CO_2 uptake for reduced catalyst samples in CO_2-TPD

Catalysts	Peak (°C)	CO_2 desorption (μmol CO_2/g)	Total CO2 uptake (μmol CO_2/g)
Cu-0	142	60.38	67.06
	562	6.68	
Cu-0.5	150	27.97	36.22
	583	8.25	
Cu-1.5	144	15.11	36.59
	640	21.48	
Cu-3.0	147	13.16	35.94
	650	22.78	

Table S2. Catalytic results from the literature for production of light olefins directly from syngas

Catalyst	T (°C)	P (bar)	H_2/CO	CO conv. (%)	CO_2 sel. (%)	Product distribution (%)		
						CH_4	$C_2^=$-$C_4^=$	C_{5+}
Fe/CNF [1]	300	20	2	88.0	42.0	13.0	52.0	18.0
ZnCrOx/MSAPO [2]	400	25	2.5	17.0	41.0	2.0	80.0	4.0
CoMn [3]	250	1	2	31.8	47.3	5.0	60.8	31.4
FeZn [4]	350	20	2.7[a]	95.1	15.8	25.6	53.32	13.36
Mn/γ-Fe2O3 [5]	320	10	1	57.1-55.1	31.7	11.7	61.2	18.2
MnO2/MSAPO [6]	400	25	2.5	8.5	41.0	2.0	79.2	<5.9%
FeBi/CNT [7]	350	10	1	78.3	47	26.1	35.2	14.1
Fe-MnK-AC [8]	320	20	1	85.0	48.0	22.7	39.4	29.7
FeMnLi [9]	320	15	2	85.6	34.6	14.2	36.7	36.1
Fe3O4@MnO2 [10]	340	15	2	91.8	37.9	12.0	37.4	45.7
FeZn [11]	340	15	2	95.9	21.8	21.4	40.9	29.2
FeMnCu [current work]	300	20	2	96.9	23.0	20.0	40.1	9.5

[a]: Feeding with CO_2

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